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# The rise of AI pricing: Trends, driving forces, and implications for firm performance ☆, ☆☆

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## ABSTRACT

We document key stylized facts about the time-series trends and cross-sectional distributions of artificial intelligence (AI)-powered pricing and study its implications for firm performance, both on average and in response to monetary policy shocks. We use the online job postings data from Lightcast to measure the adoption of AI pricing. We infer that a firm is adopting AI pricing if it posts a job that requires AI-related skills and contains the keyword “pricing.” At the aggregate level, the share of AI pricing jobs in all pricing jobs has increased more than tenfold since 2010. The rise of AI pricing jobs has been broad-based, spreading across more industries than other types of AI jobs. At the firm level, larger and more productive firms are more likely to adopt AI pricing. Firms that adopted AI pricing experienced faster growth in sales, employment, assets, and markups, and their stock returns are also more responsive to high-frequency monetary policy surprises than non-adopters. We show that these empirical observations can be rationalized by a simple model where a monopolist firm with incomplete information about its demand function invests in AI pricing to acquire information.

## 1. Introduction

Recent advances in artificial intelligence (AI) and other advanced technologies have spurred much interest in understanding their macroeconomic impacts and related policy implications. One area that has received less attention but is equally important is the

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